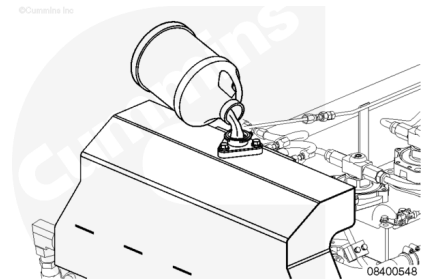


## Trainings ▾

### General Information

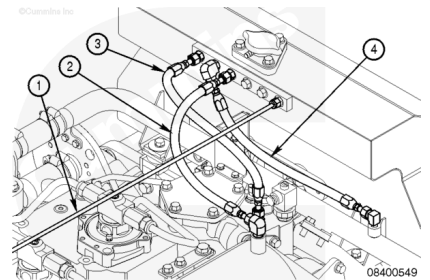
#### Marine Applications

The cooling system **must** be designed to allow the air to escape while filling the cooling system. The vent lines **must** have a continuous rise to prevent air lock and inadequate venting.



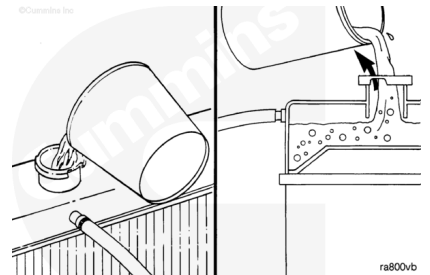
There are four vent lines attached to the expansion tank to allow the air to escape during filling of the system.

1. Turbocharger turbine housing vent
2. Wet exhaust manifold vent
3. Thermostat housing vent
4. Coolant bypass line vent.

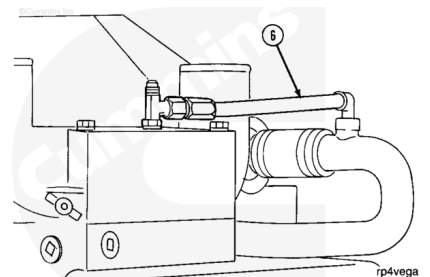


#### Conventional Aftercooling

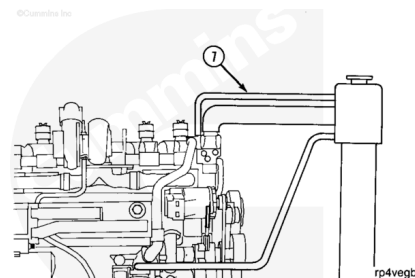
The cooling system **must** be designed to allow the air to escape while filling the cooling system.



The aftercooler vent line is routed from the aftercooler outlet tube to the top of the thermostat housing (6).



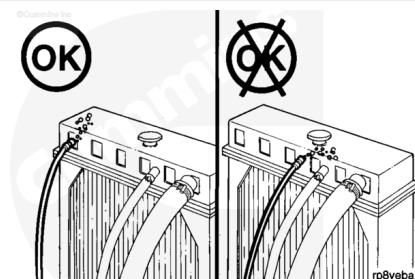
A second vent line (7) is routed from the top of the thermostat housing to the radiator top tank expansion space above the water level.



### ⚠ CAUTION ⚠

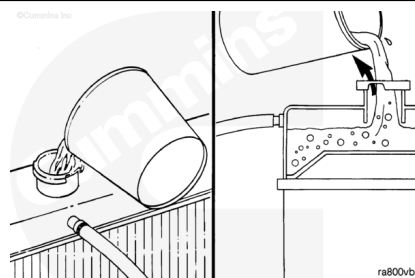
**Do not route the FILL or vent line in a manner that will allow air to be trapped in the system.**

- Route the vent line away from the makeup or FILL line.
- The vent line **must** have a continuous rise to prevent air lock and inadequate venting.

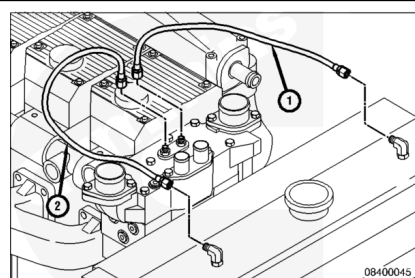


### Low Temperature Aftercooling

The cooling system **must** be designed to allow the air to escape while filling the cooling system.



The engine vent line (1) and the aftercooler vent line (2) are routed from the top of the thermostat housing to the radiator above the water level.



### ⚠ CAUTION ⚠

**Do not route the FILL or vent line in a manner that will allow air to be trapped in the system.**

- Route the vent line away from the makeup or FILL line.
- The vent line **must** have a continuous rise to prevent air lock and inadequate venting.

